

MAT 426, Class 17: Orthogonal Diagonalization

THEOREM *If A is a real symmetric matrix, then the eigenvalues of A are real.*

Proof.

THEOREM *If A is a symmetric matrix, then any two eigenvectors corresponding to distinct eigenvalues of A are orthogonal.*

Proof.

THEOREM *Let A be an $n \times n$ real matrix. Then A is symmetric if and only if it is orthogonally diagonalizable.*

Proof.

SPECTRAL THEOREM *An $n \times n$ symmetric matrix A has the following properties:*

- (a) A has n real eigenvalues, counting multiplicities*
- (b) The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.*
- (c) The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal*
- (d) A is orthogonally diagonalizable.*

Example 1. Orthogonally diagonalize the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Spectral Decomposition of A

Example 2. Find the spectral decomposition of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.